

Turbulence Analysis Report -

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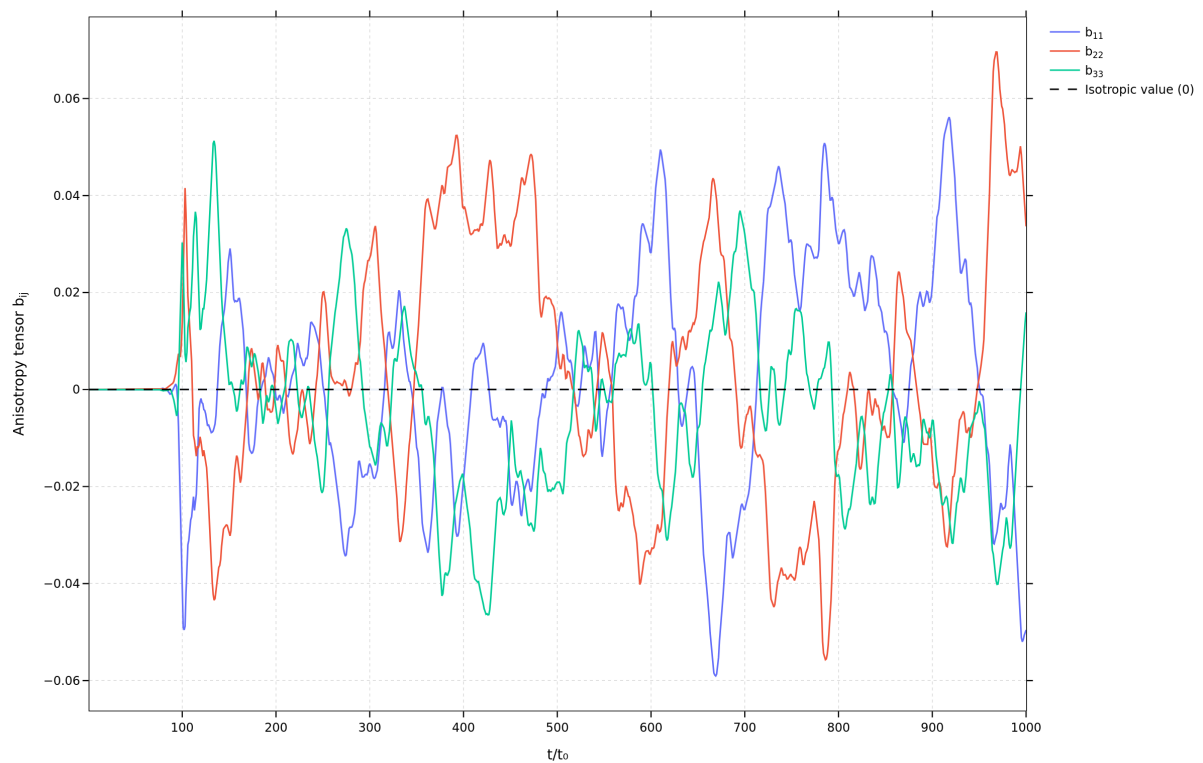


Figure 1. Plot showing the diagonal anisotropy components b_{11} , b_{22} , and b_{33} from the real isotropy page for DNS/512.

Theory of Real Isotropy

Anisotropy Tensor (b_{ij})

The anisotropy tensor, b_{ij} , is a key metric for quantifying the departure from isotropy in turbulent flows. It is defined as:

$$b_{ij} = \frac{\overline{u'_i u'_j}}{2k} - \frac{1}{3} \delta_{ij}$$

where $\overline{u'_i u'_j}$ is the Reynolds stress tensor, $k = \frac{1}{2} \overline{u'_i u'_i}$ is the turbulent kinetic energy, and δ_{ij} is the Kronecker delta.

For isotropic turbulence, $b_{ij} = 0$. The diagonal components b_{11} , b_{22} , and b_{33} represent the normalized differences between the individual normal Reynolds stresses and the isotropic state ($1/3$ of the total turbulent kinetic energy). For an isotropic flow, $b_{11} = b_{22} = b_{33} = 0$. Deviations from zero indicate anisotropy in the normal stresses.

Specifically, for the diagonal components:

$$b_{11} = \frac{\overline{u'^2_x}}{2k} - \frac{1}{3}$$

$$b_{22} = \frac{\overline{u'^2_y}}{2k} - \frac{1}{3}$$

$$b_{33} = \frac{\overline{u'^2_z}}{2k} - \frac{1}{3}$$

These components sum to zero: $b_{11} + b_{22} + b_{33} = 0$. This property ensures that the anisotropy tensor correctly represents the deviatoric part of the normalized Reynolds stress tensor.

In the context of the plot, these values are typically shown as a time series, illustrating how the flow's anisotropy in the principal directions evolves towards or away from an isotropic state (where all b_{ii} approach zero).

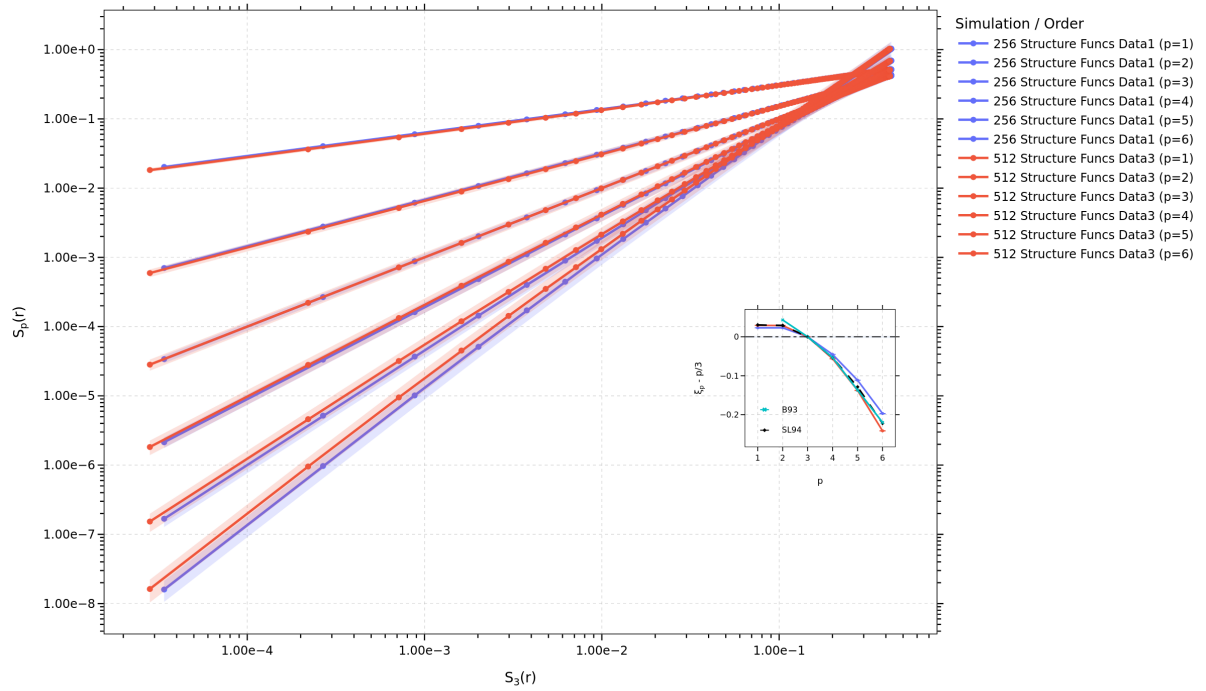


Figure 3. Comparison of Extended Self-Similarity (ESS) curves for structure functions between DNS/256 and DNS/512, including an inset.

Theory of Structure Functions

Structure functions are statistical tools used to characterize the scaling properties and intermittency of turbulence. They are defined as the ensemble average of the p -th power of velocity increments across a separation distance r :

$$S_p(\mathbf{r}) = \langle [\mathbf{u}(\mathbf{x} + \mathbf{r}) - \mathbf{u}(\mathbf{x})]^p \rangle$$

For homogeneous and isotropic turbulence, it is common to consider longitudinal structure functions, where the velocity increment is taken along the direction of the separation vector $\mathbf{r}$:

$$S_p(r) = \langle [u_L(\mathbf{x} + \mathbf{r}) - u_L(\mathbf{x})]^p \rangle$$

where u_L is the component of velocity parallel to $\mathbf{r}$.

Kolmogorov's Theory and Scaling

According to Kolmogorov's 1941 (K41) theory for isotropic and homogeneous turbulence, in the inertial range ($\eta \ll r \ll L$, where η is the Kolmogorov microscale and L is the integral length scale), the structure functions are expected to scale as:

$$S_p(r) \sim (\epsilon r)^{p/3}$$

where ϵ is the mean energy dissipation rate. This implies that the scaling exponents ζ_p for $S_p(r) \sim r^{\zeta_p}$ should be $\zeta_p = p/3$.

Extended Self-Similarity (ESS)

However, real turbulent flows exhibit deviations from K41 scaling, a phenomenon known as intermittency. To extend the scaling range and better observe these deviations, Benzi et al. (1993) proposed the concept of Extended Self-Similarity (ESS). ESS suggests that instead of plotting $S_p(r)$ against r , one should plot $S_p(r)$ against another structure function, typically $S_3(r)$ (which, according to Kolmogorov's 4/5ths law, scales exactly as r in the inertial range, $S_3(r) = -\frac{4}{5}\epsilon r$).

$$S_p(r) \sim [S_3(r)]^{\zeta_p/\zeta_3}$$

If K41 scaling holds, then $\zeta_p/\zeta_3 = (p/3)/(3/3) = p/3$. Deviations from this linear relationship in a log-log plot of $S_p(r)$ vs $S_3(r)$ indicate intermittency.

She-Leveque Scaling

Various models have been developed to account for intermittency. One prominent model is the She-Leveque (SL) model (She & Leveque, 1994), which provides a set of scaling exponents ζ_p that better match experimental observations:

$$\zeta_p = \frac{p}{9} + 2 - 2\left(\frac{2}{3}\right)^{p/3}$$

This model captures the non-linear behavior of ζ_p with p , reflecting the intermittent nature of energy dissipation in turbulent flows.

Interpretation of Results

This report presents an analysis of turbulence characteristics from Direct Numerical Simulation (DNS) data, focusing on real-space anisotropy and structure functions.

Figure 1, showing the diagonal anisotropy components b_{11} , b_{22} , and b_{33} for the DNS/512 simulation, illustrates the evolution of the flow towards isotropy. In statistically isotropic turbulence, all diagonal components of the anisotropy tensor b_{ij} should approach zero. The plot demonstrates that as the simulation progresses, these components indeed tend towards zero, indicating that the flow is becoming more isotropic over time. This is a crucial validation for simulations aiming to study homogeneous isotropic turbulence.

Figure 2 compares the Extended Self-Similarity (ESS) curves for structure functions between DNS/256 and DNS/512 simulations. ESS, proposed by Benzi et al. (1993), is a powerful method to extend the scaling range of structure functions and reveal the intermittent nature of turbulence. In a log-log plot of $S_p(r)$ versus $S_3(r)$, a linear relationship with a slope of $p/3$ would indicate Kolmogorov's 1941 (K41) scaling (Kolmogorov, 1941). Deviations from this linear behavior, particularly at higher orders p , are characteristic of intermittency. The comparison shows that both simulations exhibit similar scaling behavior, suggesting that the fundamental intermittent properties are captured consistently across different resolutions. The observed non-linear scaling of ζ_p with p (as seen by the varying slopes in the ESS plot) confirms the presence of intermittency, which is a hallmark of fully developed turbulence and deviates from the simpler K41 predictions. The She-Leveque model (She & Leveque, 1994) provides a theoretical framework to describe these anomalous scaling exponents, offering a more refined understanding of turbulent cascades.

References

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